Motor Vehicle Injurious Outcome Classification
A project by Lily Holmes

Summary

Using more than 270,000 crash records spanning over a decade, I compared logistic regression, decision tree, and naïve Bayes classifiers. Due to the human cost of serious injuries and imbalance in the outcome variable, I use recall as a criterion for model evaluation rather than accuracy alone.

The final balanced logistic regression with an adjusted probability threshold achieved a 69.2% recall, a 0.84 ROC-AUC, and an 8.1% precision on the temporally held out test set. The limitations in the design and the suboptimal precision argue against real world deployment or decision-making. Thus, this project serves as a demonstration than a public safety system.

Purpose

The primary purpose of this project was to develop a classifier to predict which motor vehicle accidents in the Denver, Colorado resulted in a harmful outcome, based on the accident features themselves, using retroactive reports.

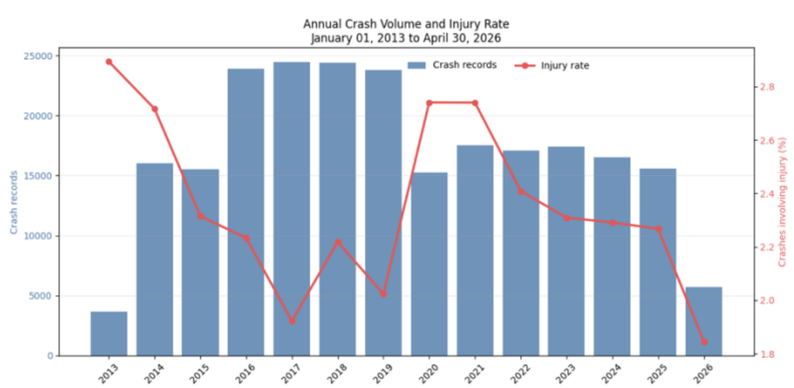
The motivation scenario was an automated system capable of flagging accidents that may require emergency medical attention in order to reduce harm and contribute to transportation safety.

Data

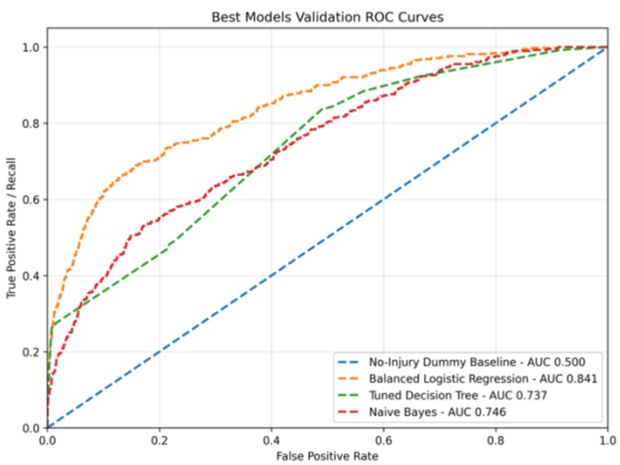
The two datasets for this project, the motor vehicle accident observations (MVA) and the city street-lines, came from the City of Denver's Open Data Repository as bulk downloaded the .csv files. The data generating process is triggered when an accident occurs and law enforcement logs information pertaining to the crash and outcome.

Between January 2013 and April 2026, there were over 270,000 observations logged. Columns in the MVA dataset included the time and location of the accident, as well as information on the road type, road condition, information on the vehicles involved, the driver action (turning, slowing, etc.), driver contributing factor (distracted, speeding etc.). I used a binary variable to indicate if there was a serious injury or fatality logged.

I created temporal, geospatial, and mechanical features to capture interaction effects. For example, taking note of hour, or nighttime, and weekend attributes. I joined the data on the city street lines data to ascertain speed limits and highway indicators. More features were created to represent size and direction difference between vehicles, as well as attributing driver actions to the corresponding vehicle size.



Methods and Metrics
I use **decision trees, logistic regression, and naïve Bayesian classifiers** compared against each other and a baseline classifier.



Due to the presence of sever class imbalance, wherein injuries are rare, I did not use accuracy alone as an evaluation metric. In this context of human life, I prioritized reducing false negatives, as it could be argued that a missed injury is more costly than a false alarm. This process included analyzing the optimal probability thresholds.

I used a temporal split to test the models on out of sample observations because for this situation, it closer simulates a real world machine learning deployment, wherein the model is trained on the past and deployed on future observations.

Results
I chose a **balanced logistic regression model with a probability threshold of .48** due to its performance on the validation set, compared to the best models of the other families.

Balanced Logistic Regression performance	
Accuracy	.825
Recall	.692
Precision	.081
F1 Score	.145
ROC_AUC	.845
PR_AUC	.229

When implemented for the singular test run, **the model identified about 69% of serious injury/fatal crashes**, but only about **8% of its positive predictions corresponded to one**. In an automatic EMS-dispatch scenario, that false-positive burden would be unacceptable. The final test therefore supports the narrower conclusion that this is a useful demonstration of rare-event modeling and honest temporal evaluation, not a system that should be deployed.

Limitations
The limitations of this design further show why this is a portfolio project rather than a deployed system. There are several omitted variables that were not available, such as seatbelt use or other roadway features. Secondly, the data collection process occurred and was completed after the outcome of a crash was known and entered, which paves for the possibility of bias in data entry. This model was predictive and not causal. There is also a lack of proof for external validity.

Project Value
This project is used to demonstrate an end to end applied machine learning work flow. This include

- Cleaning of real world, public administrative data
- Domain related feature engineering and Geospatial data integration
- Imbalanced classification and Probability threshold evaluation
- Comparison and evaluation of multiple models with a temporal split separation
- Basic automated testing